Roll No. ……

CEN-505

B. Tech. (Computer Engineering)

V Semester Examination- 2021

COMPUTER NETWORKS

Paper No.: CEN-505

Time: 3 Hours **MM: 60**

Note: Attempt all questions. Make suitable assumption(s) wherever required.

Q. 1 (a) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to four groups of customers, A, B, C and D requesting 8000, 16000, 4000 and 2000 addresses respectively. Design the sub blocks and find out how many addresses are still available after these allocations. **(6)**

CO-1

(b) Discuss the following with respect to each- physical layer, data link layer, network layer and transport layer of OSI model: **(6)**

- Name of the data unit
- Addressing Scheme used
- Device used
- Delivery of data (from and to)

OR

(b')(i) In TCP/IP protocol suite, explain how can transport layer be both connection-oriented and connectionless, when the underlying network layer protocol is connectionless only? **(3)**

(ii) Consider an IP packet with a length of 1200 bytes that includes a 20-byte IPv4 header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 150 bytes. Determine the number of fragments that the IP packet is divided into for transmission? **(3)**

Q. 2 (a) Consider a message having length, five times greater than the network can handle. So, it has been fragmented into five equal size datagrams and send to the destination. Each datagram follows a different path as given in the table: **(6)**

CO-2

Datagram	Path length	Visited switches
1	3,500 km	1,2,5
2	12,500 km	1,3,5
3	13,200 km	1,3,2,5
4	10,700 km	1,4,3,5
5	11,200 km	1,4,5

Assume the delay at each switch is 3, 10, 23, 8 and 16 msec respectively and the propagation speed is $2*10^8$m/sec. Calculate the total delay for each datagram and the order they arrive at the destination.

(b) What is the purpose of various bands in unguided transmission media? How does sky propagation differ from line-of-sight propagation? **(6)**

Q. 3 **CO-3**

(a) The following character encoding is used in a data link protocol: **(6)**
A:01001111; B:11000011; FLAG: 01111110; ESC: 11111000

Show the bit sequence transmitted (in binary) for the five-character frame: A B FLAG ESC FLAG when each of the following framing methods are used:

i) Character Count
ii) Flag bytes with byte stuffing
iii) Starting and ending flag bytes, with bit stuffing

(b) A system is generating frame at the rate of 80Mbps, while the rate of the receiver is 8Mbps for sending acknowledgement. Size of the frame and acknowledgement are 500 bytes and 50 bytes respectively. The round trip delay is 100 μs. Calculate the throughput for stop and wait ARQ. **(6)**

OR

(b') Suppose the data arrived at the receiver is FC6. Now, by using some error detecting technique, check whether the received data is error free or not. If not, then determine the original data. **(6)**

Q. 4 **CO-4**

(a) Discuss the solution to hidden station problem and exposed station problem? What is the role of inter-frame spacing in wireless LAN? **(6)**

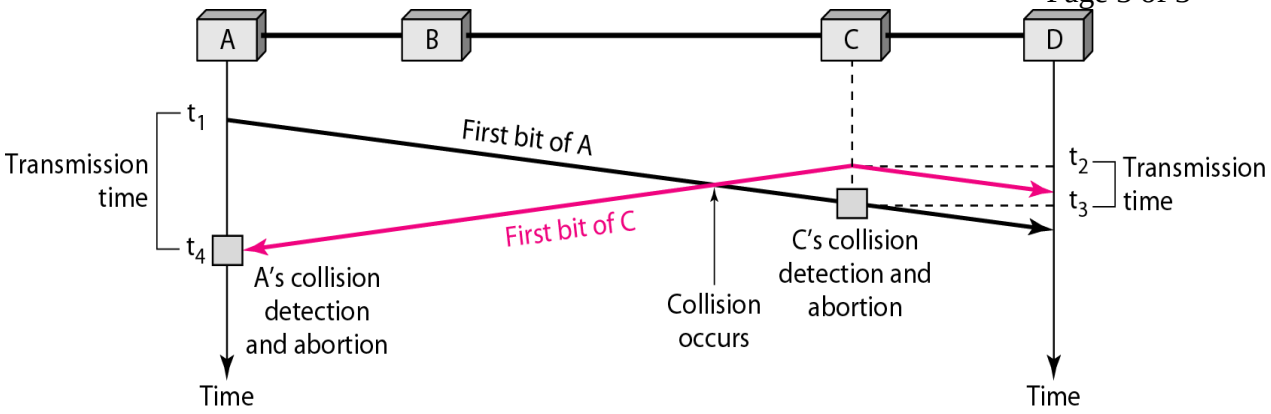
(b)(i) What is meant by vulnerable period? Show that the vulnerable time period of slotted ALOHA is half of the pure ALOHA? **(3)**

(ii) Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 500 bits. The channel transmission rate is 1 Mbps. The aggregate number of transmission across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 500 frames per second. Find the throughput of the network? **(3)**

OR

(b') In the figure given below, the data rate is 10Mbps, the distance between station A and C is 2500m, and the propagation speed is 2 x 10^8m/s. Station A starts sending a long frame at time $t_1 = 0$; station C starts sending a long frame at time $t_2 = 3\mu s$. The size of the frame is long enough to guarantee the detection of collision by both stations. Find **(6)**

i. The time when station C hears the collision (t_3)
ii. The time when station A hears the collision (t_4)
iii. The number of bits station A has sent before detecting the collision.
iv. The number of bits station C has sent before detecting the collision



Q. 5 (a) Discuss various reasons of congestion in the network? Differentiate between congestion control and flow control. Give example in order to differentiate both. **(6)**

CO-5

(b)(i) What is the benefit of inserting leaky bucket after the token bucket, explain with suitable example? **(3)**

(ii) Imagine token bucket algorithm is running for regulating the traffic at a rate of 10 Mbps. Tokens arrives at a rate of 2 Mbps. Assuming that token bucket is filled to capacity with 15Mb. Calculate the length of maximum rate burst? **(3)**

OR

(b') For the network given below, obtain the routing table of router B using link state routing. **(6)**

